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Bihar Engineering University, Patna

B.Tech End Semester Exam., 2022

Code: 110729 ***Time: 03 Hours***

Semester: VII ***Subject: ARTIFICIAL NEURAL NETWORK*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven of the following: ***[2 x 7 = 14]***

(a) Define neural network.

(b) What are dendrite?

(c) What is learning rate parameter?

(d) Mention merits of back propagation network.

(e) What is Hamming distance?

(f) Name the two types of Directional Associative Memory?

(g) What are competitive neural network?

(h) What is RBF neural network?

(i) Mention the two layer in ART 1.

(j) List the main features of RBF NN.

Q.2 ***[7+7=14]***

(a) Exemplify Linearly seperable problem.

(b) Show the mapping and structural view point of ANN.

Q.3 ***[7+7=14]***

(a) List and explain different learning laws.

(b) Discuss briefly the structure and function of a biological neuron.

Q.4 ***[7+7=14]***

(a) Mention the merits and demerits of back propagation network.

(b) Discuss briefly the practical considerations in implementation of feed-forward neural networks.

Q.5 ***[7+7=14]***

(a) What is Hop field memory? Explain briefly.

(b) Provide the training algorithm for the RBF NN with its flowchart.

Q.6 ***[7]***

Distinguish between supervisory learning and Un supervisory learning in ANN.

Q.7 ***[14]***

List the factors that affect the performance of multilayer feed-forward neural network.

Q.8 ***[14]***

How does ANN solve an XOR problem?

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Q.9 ***[14]***

How are ANN different from normal computers?

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